

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Industry based Problem-Solving

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4 – Industry based Problem-Solving	Assessment:	Assessment Tool 2 – Industry based Problem-Solving
Weightage:	40% of CIA	Total Marks:	100
CO4 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
Student Name:	JOTHIKA M	Reg. No.:	192324235
Section / Batch:	2023-B.Tech(AI & DS)	Date:	18-08-26

Code of Conduct

I Jothika M (192324235) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *Jothika M*

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	

		reasoning and insights				
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

SOLUTIONS – POINT ESTIMATION AND CONFIDENCE INTERVAL

Industry-based Problem-Solving

1. E-Commerce – Point Estimation

Question: An online retailer has 50,000 customers. A random sample of 500 customers shows that 68% made a purchase during a promotional campaign. Estimate the proportion of all customers who made a purchase.

Given: Population size $N = 50,000$; Sample size $n = 500$; Sample proportion $\hat{p} = 68\% = 0.68$.

1. Population: All 50,000 customers of the online retailer.

2. Sample: 500 randomly selected customers.

3. Parameter: Population proportion p , representing the proportion of all customers who made a purchase.

4. Point Estimate: The point estimate of the population proportion is the sample proportion.

$$\hat{p} = x / n = 0.68$$

$$\text{Therefore, } \hat{p} = 0.68 = 68\%$$

5. Interpretation: The estimated proportion of all customers who made a purchase during the promotional campaign is 68%.

Industry Interpretation: Based on the random sample, approximately 68% of the retailer's 50,000 customers are estimated to have made a purchase.

Confidence Interval

A confidence interval gives a lower and upper limit around a point estimate. The interval expresses uncertainty caused by sampling. For a 95% confidence interval, the method is designed so that, over repeated random samples, approximately 95% of intervals constructed in this way would contain the true population parameter.

Why These Concepts Matter in Industry

- They allow organizations to make decisions without measuring the entire population.
- They provide numerical evidence for planning and forecasting.
- They help quantify uncertainty instead of reporting only a single estimate.
- They support quality control, customer analytics, financial analysis, and healthcare monitoring.
- They make sample-based conclusions easier to communicate to managers and decision-makers.

2. Manufacturing – Point Estimation

Question: A factory produces electronic components. From a random sample of 100 components, the average lifetime is 1,250 hours. Determine the point estimate of the population mean lifetime.

Given: Sample size $n = 100$; Sample mean $\bar{x} = 1,250$ hours.

1. Population: All electronic components produced by the factory.

2. Sample: 100 randomly selected electronic components.

3. Parameter: Population mean lifetime μ .

4. Point Estimate: The point estimate of the population mean is the sample mean.

$$\hat{\mu} = \bar{x} = 1,250 \text{ hours}$$

5. Interpretation: The estimated average lifetime of all electronic components produced by the factory is 1,250 hours.

Industry Interpretation: Based on the sample, the factory can estimate that its electronic components have an average lifetime of approximately 1,250 hours.

3. Banking – Point Estimation

Question: A bank wants to estimate the average monthly transaction value of its customers. A sample of 200 customers has a mean transaction value of ₹18,500 and standard deviation of ₹4,200. Calculate the point estimates of the population mean and variance.

Given:

- Sample size $n = 200$;
- Sample mean $\bar{x} = ₹18,500$;
- Sample standard deviation $s = ₹4,200$.

1. Population: All customers of the bank.

2. Sample: 200 randomly selected bank customers.

3. Parameters: Population mean μ and population variance σ^2 .

4. Point Estimate of Population Mean: The sample mean is the point estimate of the population mean.

$$\hat{\mu} = \bar{x} = ₹18,500$$

5. Point Estimate of Population Variance: The sample variance is the point estimate of the population variance.

$$\hat{\sigma}^2 = s^2 = (4,200)^2 = 17,640,000 \text{ ₹}^2$$

6. Final Point Estimates:

- Population mean = ₹18,500;
- Population variance = 17,640,000 ₹².

7. Interpretation: The estimated average monthly transaction value is ₹18,500, while the estimated population variance is 17,640,000 ₹².

Industry Interpretation: The bank can use these estimates to understand the typical transaction value and the variability of customer transactions.

4. Healthcare Analytics – Point Estimation

Question: A hospital randomly selects 400 patients, of whom 52 experienced a particular side effect. Find the point estimate of the proportion of patients experiencing the side effect. Explain what the estimate means.

Given:

- Sample size $n = 400$;
- Number experiencing the side effect $x = 52$.

1. Population: All patients treated by the hospital.

2. Sample: 400 randomly selected patients.

3. Parameter: Population proportion p , representing the proportion of all patients experiencing the side effect.

4. Point Estimate: The point estimate of the population proportion is:

$$\hat{p} = x / n = 52 / 400 = 0.13$$

$$\hat{p} = 0.13 \times 100 = 13\%$$

5. Interpretation: The estimated proportion of patients experiencing the particular side effect is 13%.

Industry Interpretation: Based on the random sample, approximately 13 out of every 100 patients treated by the hospital are estimated to experience the particular side effect.

5. Food Industry – Confidence Interval

Question: A food-processing company measures the weight of 64 randomly selected packets. The mean weight is 498 g and the sample standard deviation is 12 g. Construct a 95% confidence interval for the true average packet weight.

Given:

- Sample size $n = 64$;
- Sample mean $\bar{x} = 498$ g;
- Sample standard deviation $s = 12$ g;
- Confidence level = 95%.

1. Population: All packets produced by the food-processing company.

2. Sample: 64 randomly selected packets.

3. Parameter: Population mean packet weight μ .

4. Degrees of Freedom: $df = n - 1 = 64 - 1 = 63$.

5. Critical Value: For a 95% confidence interval with 63 degrees of freedom, $t_{0.025,63} \approx 2.000$.

6. Standard Error: $SE = s / \sqrt{n} = 12 / \sqrt{64} = 12 / 8 = 1.5$ g.

$$CI = \bar{x} \pm t(\alpha/2) \times (s / \sqrt{n})$$

$$CI = 498 \pm (2.000)(1.5)$$

$$CI = 498 \pm 3$$

7. Lower Limit: $498 - 3 = 495$ g.

8. Upper Limit: $498 + 3 = 501$ g.

$$\text{95\% Confidence Interval} = (495 \text{ g}, 501 \text{ g})$$

9. Final Answer: $495 \text{ g} < \mu < 501 \text{ g}$.

10. Interpretation: We are 95% confident that the true average weight of all packets produced by the company lies between 495 g and 501 g.

Industry Interpretation: The company's production process is estimated to have a true average packet weight in the range of 495 g to 501 g. This interval can help the company assess whether its average packet weight is close to the desired target of 500 g.

Answers

No.	Industry	Type	Final Answer
1	E-Commerce	Point Estimation	68%
2	Manufacturing	Point Estimation	1,250 hours
3	Banking	Point Estimation	Mean = ₹18,500; Variance = 17,640,000 ₹ ²
4	Healthcare Analytics	Point Estimation	13%
5	Food Industry	Confidence Interval	495 g to 501 g